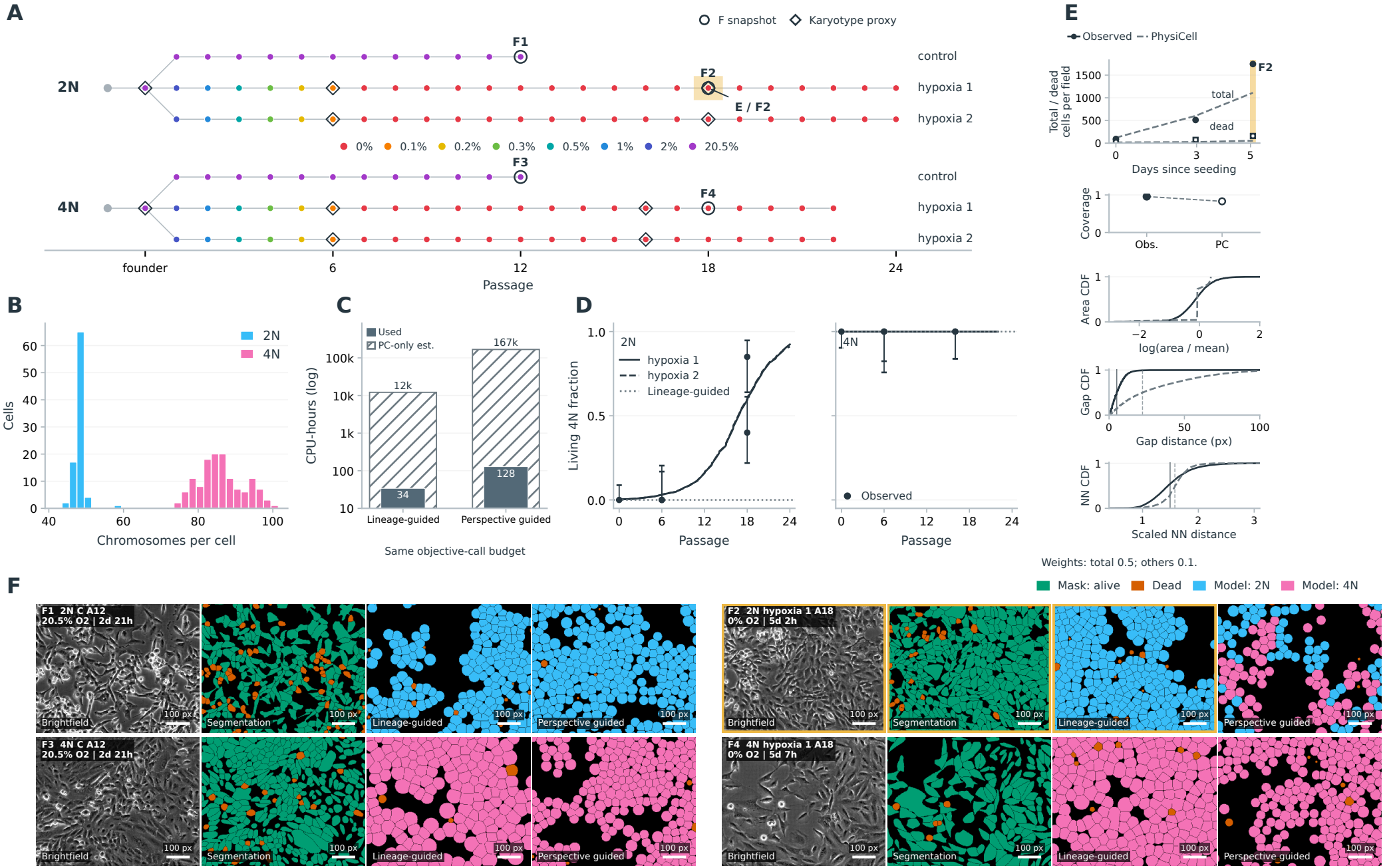

A
F snapshot
Karyotype proxy
F1
F2
E / F2
F3
F4
2N
4N
control
hypoxia 1
hypoxia 2
0%
0.1%
0.2%
0.3%
0.5%
1%
2%
20.5%
founder
6
12
18
24
Passage
B
2N
4N
Cells
Chromosomes per cell
C
Used
PC-only est.
CPU-hours (log)
12k
167k
34
128
Lineage-guided
Perspective guided
Same objective-call budget
D
2N
hypoxia 1
hypoxia 2
Lineage-guided
Living 4N fraction
Passage
4N
Observed
E
Observed
PhysiCell
Total / dead cells per field
total
dead
Days since seeding
Coverage
Obs.
PC
Area CDF
log(area / mean)
Gap CDF
Gap distance (px)
NN CDF
Scaled NN distance
Weights: total 0.5; others 0.1.
Mask: alive
Dead
Model: 2N
Model: 4N
F
F1 2N C A12
20.5% O2 | 2d 21h
F2 2N hypoxia 1 A18
0% O2 | 5d 2h
F3 4N C A12
20.5% O2 | 2d 21h
F4 4N hypoxia 1 A18
0% O2 | 5d 7h
Brightfield
Segmentation
Lineage-guided
Perspective guided
100 px